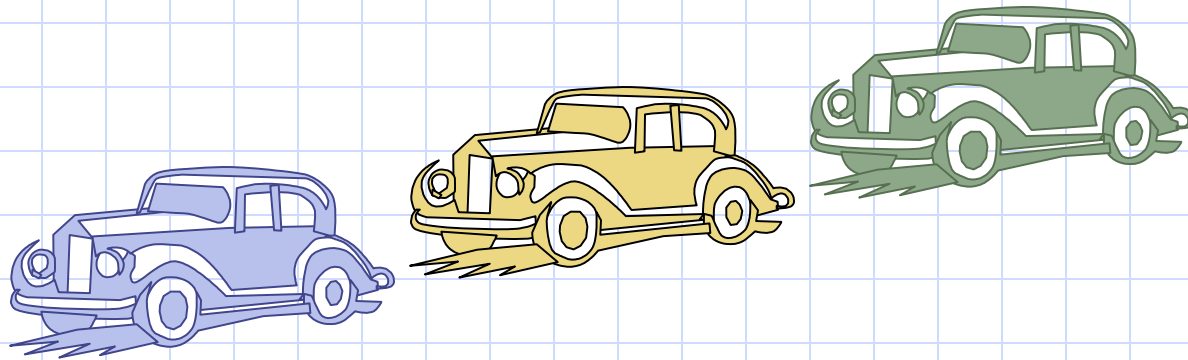


Queues



The Queue ADT

- The **Queue** ADT stores arbitrary objects
- Insertions and deletions follow the first-in first-out scheme
- Insertions are at the rear of the queue and removals are at the front of the queue
- Main queue operations:
 - **enqueue**(object): inserts an element at the end of the queue
 - object **dequeue**(): removes and returns the element at the front of the queue
- Auxiliary queue operations:
 - object **first**(): returns the element at the front without removing it
 - integer **len**(): returns the number of elements stored
 - boolean **is_empty**(): indicates whether no elements are stored
- Exceptions
 - Attempting the execution of dequeue or front on an empty queue throws an **EmptyQueueException**

Example

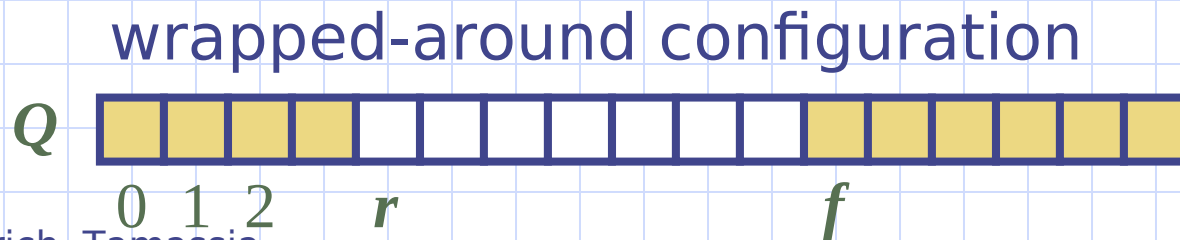
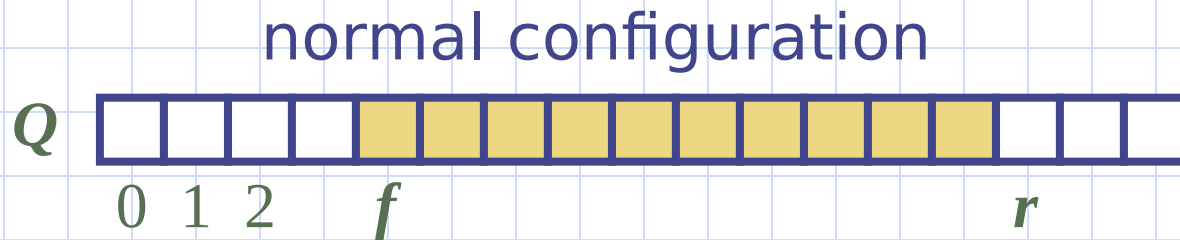
Operation	Return Value	first $\leftarrow$ Q $\leftarrow$ last
Q.enqueue(5)	—	[5]
Q.enqueue(3)	—	[5, 3]
len(Q)	2	[5, 3]
Q.dequeue()	5	[3]
Q.is_empty()	False	[3]
Q.dequeue()	3	[]
Q.is_empty()	True	[]
Q.dequeue()	“error”	[]
Q.enqueue(7)	—	[7]
Q.enqueue(9)	—	[7, 9]
Q.first()	7	[7, 9]
Q.enqueue(4)	—	[7, 9, 4]
len(Q)	3	[7, 9, 4]
Q.dequeue()	7	[9, 4]

Applications of Queues

- ❑ Direct applications
 - Waiting lists, bureaucracy
 - Access to shared resources (e.g., printer)
 - Multiprogramming
- ❑ Indirect applications
 - Auxiliary data structure for algorithms
 - Component of other data structures

Array-based Queue

- Use an array of size N in a circular fashion
- Two variables keep track of the front and rear
 - f index of the front element
 - r index immediately past the rear element
- Array location r is kept empty



Queue Operations

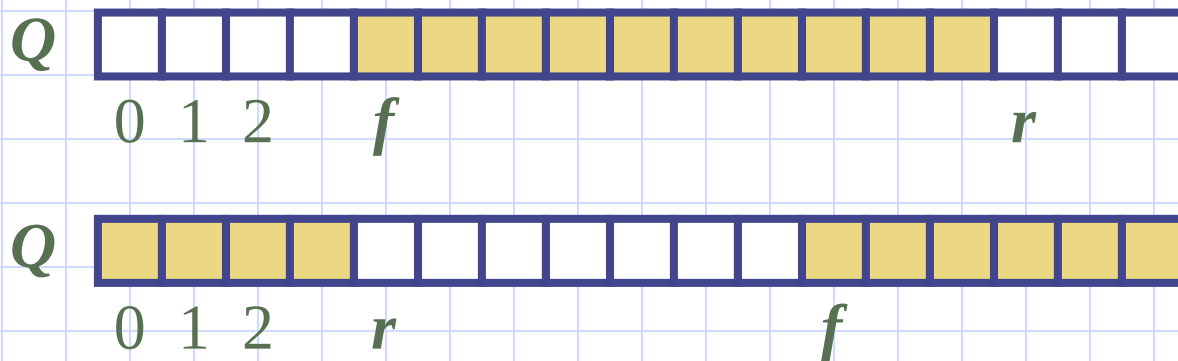
- We use the modulo operator (remainder of division)

Algorithm *size()*

return $(N - f + r) \bmod N$

Algorithm *isEmpty()*

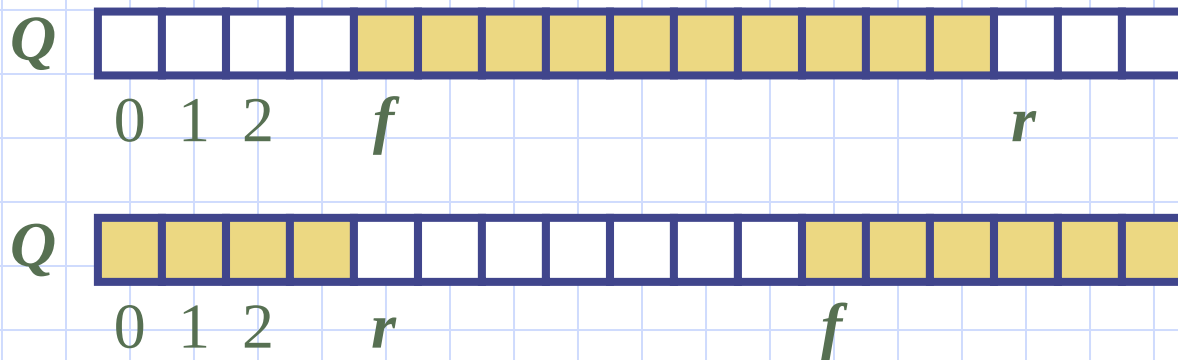
return $(f = r)$



Queue Operations (cont.)

- Operation enqueue throws an exception if the array is full
- This exception is implementation-dependent

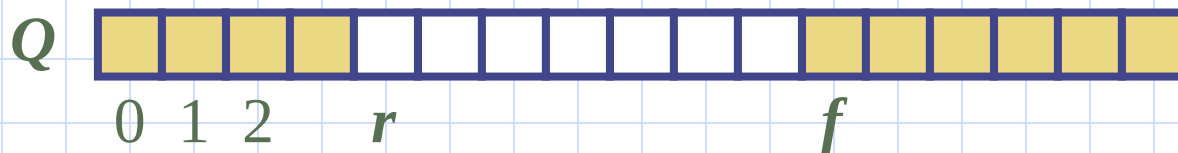
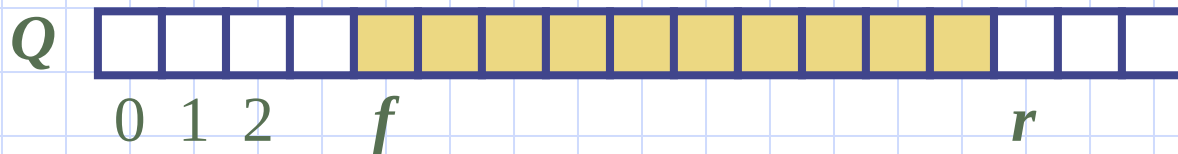
```
Algorithm enqueue(o)  
  if  $size() = N - 1$  then  
    throw FullQueueException  
  else  
     $Q[r] \leftarrow o$   
     $r \leftarrow (r + 1) \bmod N$ 
```



Queue Operations (cont.)

- ❑ Operation `dequeue` throws an exception if the queue is empty
- ❑ This exception is specified in the queue ADT

```
Algorithm dequeue()  
  if isEmpty() then  
    throw EmptyQueueException  
  else  
     $o \leftarrow Q[f]$   
     $f \leftarrow (f + 1) \bmod N$   
  return  $o$ 
```



Queue in Python

- Use the following three instance variables:
 - `_data`: is a reference to a list instance with a fixed capacity.
 - `_size`: is an integer representing the current number of elements stored in the queue (as opposed to the length of the data list).
 - `_front`: is an integer that represents the index within data of the first element of the queue (assuming the queue is not empty).

Queue in Python, Beginning

```
1 class ArrayQueue:
2     """FIFO queue implementation using a Python list as underlying storage."""
3     DEFAULT_CAPACITY = 10          # moderate capacity for all new queues
4
5     def __init__(self):
6         """Create an empty queue."""
7         self._data = [None] * ArrayQueue.DEFAULT_CAPACITY
8         self._size = 0
9         self._front = 0
10
11    def __len__(self):
12        """Return the number of elements in the queue."""
13        return self._size
14
15    def is_empty(self):
16        """Return True if the queue is empty."""
17        return self._size == 0
18
```

```
19    def first(self):
20        """Return (but do not remove) the element at the front of the queue.
21
22        Raise Empty exception if the queue is empty.
23        """
24        if self.is_empty():
25            raise Empty('Queue is empty')
26        return self._data[self._front]
27
28    def dequeue(self):
29        """Remove and return the first element of the queue (i.e., FIFO).
30
31        Raise Empty exception if the queue is empty.
32        """
33        if self.is_empty():
34            raise Empty('Queue is empty')
35        answer = self._data[self._front]
36        self._data[self._front] = None          # help garbage collection
37        self._front = (self._front + 1) % len(self._data)
38        self._size -= 1
39        return answer
```

Queue in Python, Continued

```
40 def enqueue(self, e):
41     """ Add an element to the back of queue."""
42     if self._size == len(self._data):
43         self._resize(2 * len(self._data))    # double the array size
44     avail = (self._front + self._size) % len(self._data)
45     self._data[avail] = e
46     self._size += 1
47
48 def _resize(self, cap):                      # we assume cap >= len(self)
49     """ Resize to a new list of capacity >= len(self)."""
50     old = self._data                         # keep track of existing list
51     self._data = [None] * cap               # allocate list with new capacity
52     walk = self._front
53     for k in range(self._size):              # only consider existing elements
54         self._data[k] = old[walk]            # intentionally shift indices
55         walk = (1 + walk) % len(old)         # use old size as modulus
56     self._front = 0                          # front has been realigned
```

Application: Round Robin Schedulers

- We can implement a round robin scheduler using a queue Q by repeatedly performing the following steps:
 1. $e = Q.dequeue()$
 2. Service element e
 3. $Q.enqueue(e)$

